Using Calculators to Solve Equations

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Using Calculators to Solve Equations

How do I find points of intersection using my graphic display calculator?

- You can use your **graphic display calculator** to draw **two graphs** on the **same axes**, $y = f_1(x)$ and $y = f_2(x)$, then find the **coordinates** of the **points of intersection**
  - These are the points where the two graphs meet
- First draw the graph of $y = f_1(x)$ on your graphic display calculator
  - Read the revision note on Using Calculators to Sketch Graphs to see how
- Then add a **second graph**, $y = f_2(x)$, on to the same axes of the graph above
  - You may find the **tab** button is a shortcut for adding a second graph
- Now navigate to "**analyze graph**" and select "**intersection**"
  - On some models, press **G-Solv** and then **INTSECT** instead
  - This finds the **coordinates** of the **points of intersection**
    - You may have to click before and after the intersection to help the calculator search for it
    - You may have to use the left and right buttons to select which intersection point you are finding

How do I solve equations using my graphic display calculator?

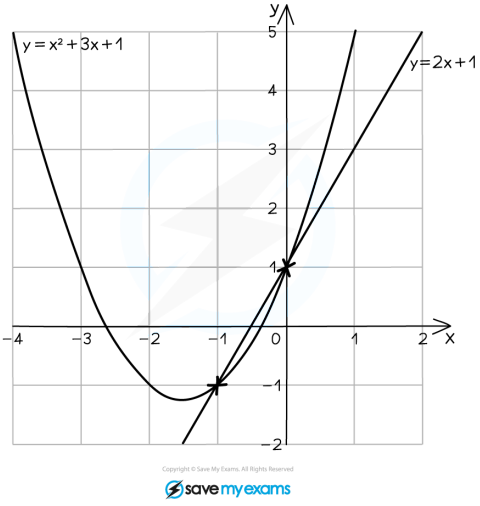
You can **solve equations** of the form $f_1(x) = f_2(x)$ using a **graphic display calculator**

- $f_1(x)$ is the **left-hand side** of the equation, $f_2(x)$ is the **right-hand side**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Help me understand

- For example, to solve $x^2 + 3x + 1 = 2x + 1$ using a graphic display calculator
  - draw $y = x^2 + 3x + 1$ on your calculator and add on $y = 2x + 1$
  - Find the two points of intersection, which are (-1, -1) and (0, 1)
  - The solutions to $x^2 + 3x + 1 = 2x + 1$ are the ***x*-coordinates** of these points
  - so $x = -1$ or $x = 0$ are the solutions



What situations should I be familiar with?

- Whilst you may be asked to solve any equation of the form $f_1(x) = f_2(x)$, there are a few familiar situations that you should recognise
- Solving a **function equal to zero**: $f_1(x) = 0$
  - For example, to solve $x^2 + 3x + 1 = 0$
    - Using above, the solutions are the *x*-coordinates of the points of intersection of $y = x^2 + 3x + 1$ and $y = 0$ (since $f_2(x) = 0$)
    - You need to recognise $y = 0$ as the ***x*-axis**, so the **solutions are the *x*-intercepts** of $y = x^2 + 3x + 1$
- Solving a **function equal to a constant** (number): $f_1(x) = k$
  - For example, to solve $x^2 + 3x + 1 = 1$
    - The solutions are the *x*-coordinates of the points of intersection of $y = x^2 + 3x + 1$ and $y = 1$ (since $f_2(x) = 1$)
    - You need to recognise $y = 1$ as a **horizontal line** that cuts the *y*-axis at 1
- To solve a **different equation** like $x^2 + 3x - 4 = 0$ using the **original graph** $y = x^2 + 3x + 1$ you need to **rearrange the different equation** to get "graph = ..."
  - **Add / subtract terms** to both sides
    - For example, add 5 to both sides of $x^2 + 3x - 4 = 0$

Is this method the same as using the equation solver on my graphic display calculator?

- The method above for solving equations of the form $f_1(x) = f_2(x)$ using the x-coordinate of intersection is called a **graphical method**
- However, the equation $f_1(x) = f_2(x)$ can be rearranged by **bringing all terms to one side**
  - $f_1(x) - f_2(x) = 0$ (or $f_2(x) - f_1(x) = 0$)
    - E.g. $x^2 + 3x + 1 = 2x + 1$ becomes $x^2 + x = 0$
  - If this forms a **quadratic or cubic equation**, you can use the **equation solver** on your calculator
    - This is no longer a graphical method but now an **algebraic method**
    - Look for the option for **solving a polynomial**
  - Select the **degree** (or **order**) of the equation you are solving
    - When solving a **quadratic**, the **degree** (or **order**) is **2**
    - When solving a **cubic**, the **degree** (or **order**) is **3**
- Type in the **coefficients** of the equation you are solving
  - This is usually in the format $a_2x^2 + a_1x + a_0 = 0$ (or $ax^2 + bx + c = 0$)
  - Then press **enter** (or **solve**) and view the **solutions**
    - This will give the **same solutions** as those from the **graphical method**

Examiner Tips and Tricks

If asked to solve an equation using a **graphical method**, you must do it by finding **points of intersections** between two graphs (not by using the equation solver).

Worked Example

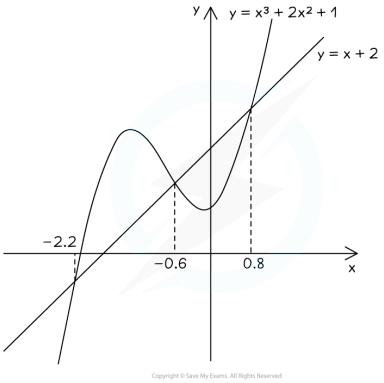
Draw the graph of $y = x^3 + 2x^2 + 1$ on your calculator.

Use a graphical method to solve the equation $x^3 + 2x^3 + 1 = x + 2$, giving your answers correct to 1 decimal place.

Start a new graph on your calculator and enter $f_1(x) = x^3 + 2x^2 + 1$ to generate the graph
Add on the second graph of $f_2(x) = x + 2$

Select the option to "analyze" the graph (this may be labelled as G-Solv)

The solutions to the equation are the x-coordinates of the points of intersection



$x = -2.2, x = -0.6$ or $x = 0.8$ to 1 d.p.

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